# Samyog GC

Amritnagar, Kirtipur, Kathmandu, Nepal
Bachelor in Computer Engineering (Final Year)

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## Professional Summary


Final-year Bachelor's student in Computer Engineering with completed projects in CNN and Random Forest for flood susceptibility mapping. Currently engaged in landfill suitability mapping using machine learning segmentation techniques. Proficient in Python, OpenGL, and QGIS, with relevant coursework in data structures, algorithms, and artificial intelligence. Motivated to deepen expertise and contribute to machine learning and data engineering through a fellowship opportunity.


## Education

**Bachelor in Computer Engineering**
*Advanced College of Engineering and Management, Tribhuvan University, Kathmandu, Nepal* 2021 – Present

**Higher Secondary Education (Science, PCM & CS)**
*Kathmandu Bernhardt College, Kathmandu, Nepal* 2019 – 2021

**Secondary Education (Grade 10)**
*Pushpasadan Boarding Secondary School, Kirtipur, Kathmandu, Nepal* 2019

## Projects

**Landfill Suitability Mapping using Image Segmentation (Ongoing)**
*Major project: Using machine learning segmentation models to identify suitable landfill sites for waste generated by Kathmandu. Integrates geospatial and environmental data using Python and QGIS.*

**Pluvial Flood Susceptibility Mapping and Risk Assessment Using Deep Learning**
*Minor project: Developed models using CNN and Random Forest to map and assess pluvial flood susceptibility and risk in Kathmandu Valley. Utilized geospatial datasets and Python-based ML frameworks like TensorFlow and scikit-learn.*

**3D World Rendering using OpenGL**
*Computer Graphics project: Created an interactive 3D cube-based world with user navigation using Python and OpenGL.*

**Flappy Bird Clone**
*Object-Oriented Programming project: Developed a Flappy Bird game clone applying object-oriented programming principles in C++.*

## Other Relevant Experience

**Participant - Inter-college Hackathon** Won Best Idea Award

**Participant - Technorian Line-Following Robot Competition**

## Technical Skills

- **Programming Languages:** C/C++, Python, VHDL, Assembly (8085, 8086), HTML/CSS, MATLAB
- **Machine Learning:** CNN, UNet, Artificial Neural Networks (ANNs), K-Nearest Neighbors (KNNs), Random Forest, Image Segmentation
- **Tools & Technologies:** TensorFlow, OpenCV, QGIS, GDAL, Rasterio, scikit-learn, NumPy, Pandas, LaTeX, Git/GitHub, Jupyter Notebooks, Visual Studio Code, Weka
- **Languages:** English (Fluent), Nepali (Native), Hindi (Conversational)

## Interests & Hobbies

Machine Learning, Football, Games, Art, Game Design, Tinkering